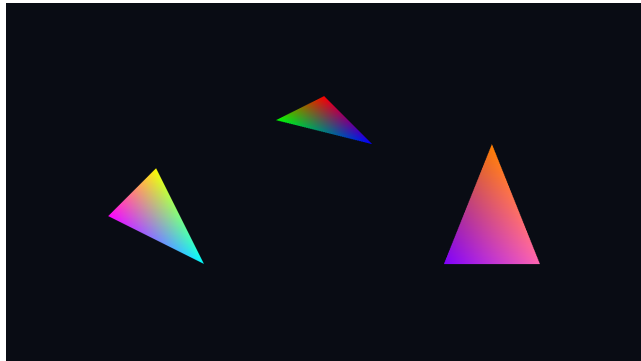


Visual Computing - Assignment 2

Pierre CHHIENG

Task 1

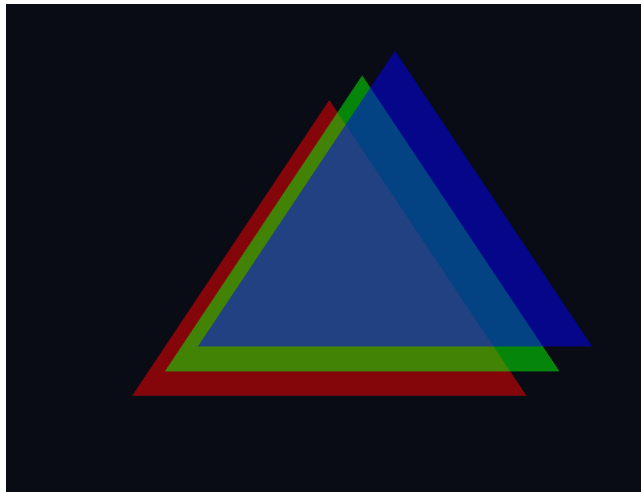
b)



When OpenGL renders a triangle, it interpolates the vertex colors across the surface of the triangle. For each fragment inside the triangle, it computes a weighted average of the colors of the three vertices, based on the fragment's position.

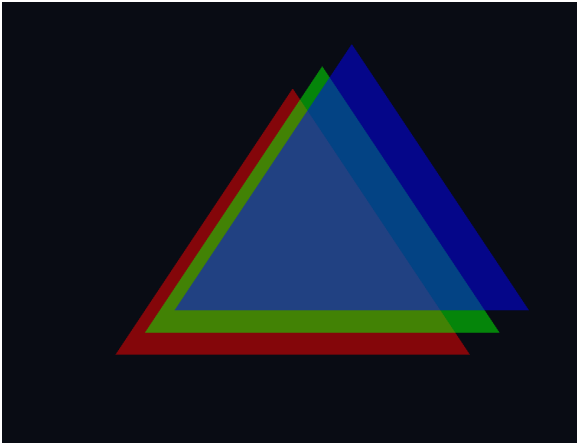
Task 2

a)

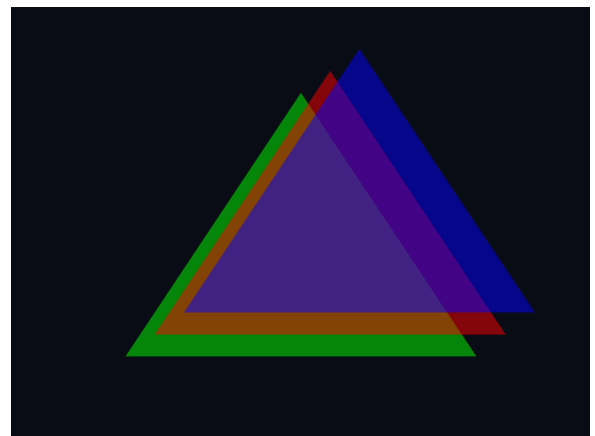


b)

i)



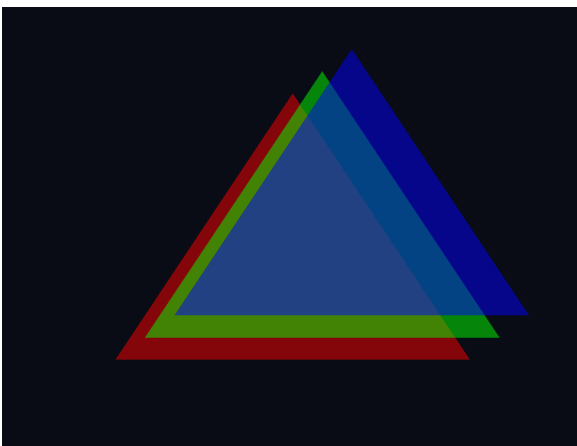
Original



Swap Red and Green

When the colors of different triangles were exchanged, the blended color in the overlapping region also changed. This occurs because the blending operation combines the source fragment's color with the existing framebuffer color, weighted by the source alpha value.

ii)



Original



Swapped the red and green triangles' depth values

When the depth values of the triangles were changed, the triangle that appeared in front of the others was also changed. Then, the blended color of the overlapping region looked different, because OpenGL first checks the depth buffer to decide which fragments are in front and which are behind. The triangle with the smallest z-value becomes the “source” in the blending calculation, so swapping z-coordinates effectively changes the blending order.

Task 3

b)

- a: Controls scaling along the x-axis. values greater than 1 stretch the object horizontally, values between 0 and 1 compress it, and negative values both scale and mirror the x-axis.
- e: Similarly, e scales the y-axis. Values greater than 1 stretch vertically, values between 0 and 1 compress, and negative values mirror along the y-axis.
- c: Translates the object along the x-axis in homogeneous coordinates, shifting the entire scene left or right.
- f: Translates the object along the y-axis, moving the scene up or down.
- b and d: Produce shear transformations. b shears x-coordinates based on y, slanting the shape sideways. d shears y-coordinates based on x, tilting horizontal lines vertically.

c)

A rotation in two dimensions requires a specific combination of values in the 2×2 submatrix. Changing only one entry cannot preserve lengths and angles, so it cannot result in a rotation.

Task 4

Keybinds

For camera movement, the following keybinds were used:

W / S: move forward and backward along the z-axis

A / D: move left and right along the x-axis

Space / Left Shift: move up and down along the y-axis

For camera rotation, the following keybinds were used:

Arrow Up / Down: pitch rotation (looking up and down)

Arrow Left / Right: yaw rotation (looking left and right)